Experiment No. 6

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AIM: To design and implement an automatic shelter system that detects rainfall using a rain sensor and responds by activating a servo motor to cover or uncover the shelter using an Arduino Uno microcontroller.

Introduction: In many outdoor applications such as plant nurseries, bike stands, or open seating areas, it is important to protect objects or people from unexpected rainfall. Manual intervention is not always possible or convenient. This project aims to automate the process of shelter deployment using a rain-sensing mechanism.

The system uses a **rain sensor module** to detect rainfall and an **Arduino Nano** to process the signal. Once rain is detected, the Arduino sends a signal to a **servo motor**, which activates a mechanical system to **close the shelter automatically**. When the rain stops, the shelter reopens. The project is powered by a **12V DC power supply** and involves additional components like wires, connectors, and structural materials to support the shelter mechanism.

This smart automation project demonstrates the integration of sensors and actuators with microcontrollers to create efficient, real-world solutions for weather-responsive systems.

Component Required:

- **Microcontroller: Arduino Nano**

Function:
Acts as the brain of the system.

Working:

- It takes input from the **Rain Sensor Module**.
- Processes the data to check if it's raining.

- Based on that, it sends signals to the **Servo Motor** to either open or close the shelter.
- Controls timing, decision-making, and signal flow between all connected components.

◇ Rain Sensor Module

Function:
Detects the presence of rain.

Working:

- It has a **rain detection board** that senses water droplets.
- When rain touches the sensor, it changes the electrical resistance.
- The sensor sends a **digital (rain detected) or analog (how much rain)** signal to the Arduino.
- Arduino reads this and triggers the shelter to close.

◇ Servo Motor

Function:
Physically drives the movement of the shelter (open/close).

Working:

- Receives **PWM (Pulse Width Modulation)** signal from Arduino.
- Rotates to a set angle (like 0°, 90°, or 180°) depending on command.
- For example:
    - If no rain → rotate to **open** the shelter.
    - If rain detected → rotate to **close** the shelter.
- Provides precise and controlled movement ideal for the project.

◇ Power Supply: 12V DC Adapter

Function:
Provides power to the entire system.

Working:

- Supplies 12V power to the circuit.
- The power is often regulated to 5V for the Arduino and other components (can be through a voltage regulator or onboard power module).
- Ensures consistent voltage for reliable functioning.

◇ Miscellaneous: Wires, Connectors, and Structure

Function:
Support system and connectivity.

Working:

- **Wires and connectors** transmit signals and power between modules.
- **Structural materials** (like acrylic/plastic/metal frames) hold the servo and shelter in place.
- Ensure stable and functional construction of the rain-sensing shelter mechanism.

Connections:

Component	**Pin on Component**	**Connected To (Arduino Nano Pin)**	**Purpose**
Rain Sensor Module	VCC	5V	To power the sensor
	GND	GND	Common ground
	D0 (Digital Output)	D2	To send rain detection signal
Servo Motor	Signal (Orange/Yellow wire)	D9	PWM signal to control motor angle

	VCC (Red wire)	5V (or external 5V source)	Power supply to servo
	GND (Brown/Black wire)	GND	Ground connection
Power Supply	+12V	External power input (via adapter)	Powers the full system
	GND	GND	Ground shared across system
Component	**Pin on Component**	**Connected To (Arduino Nano Pin)**	**Purpose**
Rain Sensor Module	VCC	5V	To power the sensor
	GND	GND	Common ground

Working Principle:

The **Rain Sensing Automatic Shelter System** operates on the principle of detecting moisture (rain) using a **rain sensor module** and triggering a **servo motor** through an **Arduino Nano** to open or close a shelter.

When rainwater falls on the sensor's surface, the **sensor detects the change in resistance** due to the presence of water droplets. This change is converted into a **digital signal** (LOW), which is sent to the Arduino Nano. The Arduino processes this input and sends a **PWM signal** to the servo motor, instructing it to **rotate and close the shelter** to protect the area beneath.

Once the sensor no longer detects rain (surface dries), the digital output returns to HIGH. The Arduino then sends another PWM signal to **reverse the servo motor's position**, **reopening the shelter**.

This system provides an automatic response to environmental changes (rain), offering protection without human intervention.

Software Implementation:

The software for this project was written using the Arduino IDE

This simple loop-based logic provides real-time feedback and control over the lock mechanism

Arduino Code:

```
#include <Servo.h>

// Define the pin for the rain sensor

const int rainSensorPin = A0; // You can change this pin as per your wiring

// Define the pin for the servo motor

const int servoPin = 8; // You can change this pin as per your wiring

Servo myServo;

void setup() {

  // Set up the rain sensor pin as an input

  pinMode(rainSensorPin, INPUT);

  // Set up the servo motor

  myServo.attach(servoPin);

}

void loop() {

  // Read the state of the rain sensor

  int rainState = digitalRead(rainSensorPin);

  // If the rain sensor detects rain (LOW), rotate the servo motor continuously

  if (rainState == LOW) {

    myServo.write(180); // Rotate the servo motor to 90 degrees

  } else {

    // If no rain is detected, stop the servo motor

    myServo.write(0); // Rotate the servo motor to 0 degrees

  }

  // Add a small delay for stability
```

```
  delay(100);
}
```

Communication:

Component	**Communication Type**	**Protocol/Method**	**Description**
Rain Sensor Module	Digital Communication	**Digital I/O (ON/OFF Signal)**	Sends a HIGH (no rain) or LOW (rain detected) signal to the Arduino via D0 pin.
Servo Motor	PWM (Pulse Width Modulation)	**PWM Signal**	Arduino sends a PWM signal to control the angle/position of the servo motor.
Arduino Nano	GPIO	**General Purpose I/O**	Acts as the controller, reading digital signals and outputting PWM.

Conclusion:

The Rain Sensing Automatic Shelter System successfully demonstrates the use of sensors and microcontrollers to automate weather-responsive actions. By detecting rain and activating a servo motor to control a shelter, the project showcases practical implementation of embedded systems. It reduces the need for manual intervention, making it ideal for outdoor environments. This smart solution reflects how simple electronics can enhance convenience, safety, and functionality in everyday life.